WORKED SOLUTION

STEP I 1993, Question 13

Mechanics • Work and Power

STEP I 1993, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

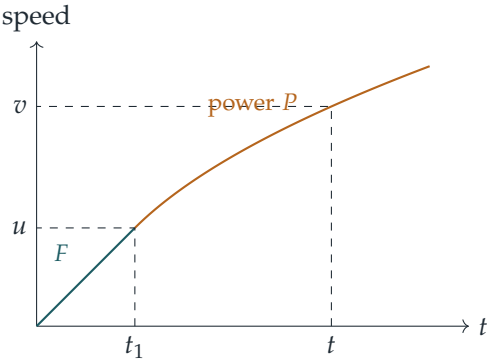
step.maths.org/questions/93-s1-q13

The key idea

The motion has two phases, and each needs its own form of Newton's second law. While the force is constant the acceleration is constant, so the *suvat* equations do everything. Once the engine works at constant power, the driving force is P/v, which changes with the speed; the trick is to write the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}t}$ (for the time) or $v\frac{\mathrm{d}v}{\mathrm{d}x}$ (for the distance) so that the variables separate.

Solution

Model the engine and train together as a particle of mass M moving along a straight horizontal track with no resistance. The only horizontal force is the tractive force (weight and normal reaction are vertical and cancel). The figure sketches the speed against time: a straight line up to speed u, then a curve whose slope falls as the speed rises.



Phase 1: constant force

With constant force F the acceleration is F/M. Starting from rest, the speed reaches u at time

$$t_1 = \frac{Mu}{F} = \frac{Mu^2}{P},$$

using $F = P/u$. By $v^2 = u^2 + 2as$ (here from 0 to u), the distance covered is

$$s_1 = \frac{u^2}{2F/M} = \frac{Mu^2}{2F} = \frac{Mu^3}{2P}.$$

Phase 2: constant power

Power is force times velocity, so at speed w the tractive force is P/w. Note that at $w = u$ this is $P/u = F$, so the force does not jump when the engine changes mode. Newton's second law gives

$$M\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{P}{w}.$$

Time. Separating the variables, $Mw\,\mathrm{d}w = P\,\mathrm{d}t$. Integrating from speed u at time t_1 to speed v at time t,

$$\frac{M}{2}\left(v^2 - u^2\right) = P\,(t - t_1).$$

(This is just the work–energy principle: the engine does work $P(t - t_1)$, which all becomes kinetic energy.) So

$$t = \frac{Mu^2}{P} + \frac{M}{2P}\left(v^2 - u^2\right) = \frac{M}{2P}\left(2u^2 + v^2 - u^2\right) = \frac{M}{2P}\left(u^2 + v^2\right),$$

as required. Since $v \geq u$, the speed v is reached in phase 2 (or exactly at its start), which is why we may use phase 2's equation.

Ask yourself For the distance we need x, not t, as the independent variable. How do we rewrite the acceleration?

By the chain rule, $\dfrac{\mathrm{d}w}{\mathrm{d}t} = \dfrac{\mathrm{d}w}{\mathrm{d}x}\dfrac{\mathrm{d}x}{\mathrm{d}t} = w\dfrac{\mathrm{d}w}{\mathrm{d}x}$. So the equation of motion becomes

$$Mw\frac{\mathrm{d}w}{\mathrm{d}x} = \frac{P}{w}, \qquad \text{that is,} \qquad Mw^2\,\mathrm{d}w = P\,\mathrm{d}x.$$

Distance. Integrating from $w = u$ to $w = v$, the distance s_2 travelled in phase 2 satisfies

$$\frac{M}{3}\left(v^3 - u^3\right) = P\,s_2, \qquad s_2 = \frac{M}{3P}\left(v^3 - u^3\right).$$

The total distance is therefore

$$\begin{aligned} s_1 + s_2 &= \frac{Mu^3}{2P} + \frac{M}{3P}\left(v^3 - u^3\right) \\ &= \frac{M}{6P}\left(3u^3 + 2v^3 - 2u^3\right) \\ &= \frac{M}{6P}\left(2v^3 + u^3\right). \end{aligned}$$

A misprint in the paper. The printed answer has u^2 in place of u^3. That can't be right: Mv^3/P is a length (power has units $\mathrm{kg\,m^2\,s^{-3}}$, and Mv^3 has units $\mathrm{kg\,m^3\,s^{-3}}$), whereas Mu^2/P is a time, so the u term must also be a cube for the two terms to be added. The correct distance is $\dfrac{M}{6P}\left(2v^3 + u^3\right)$, which is what we have shown.

Checks

- Take $v = u$: the time formula gives $Mu^2/P = Mu/F$ and the distance gives $Mu^3/(2P) = Mu^2/(2F)$, exactly the phase 1 results, as they must be.
- Units: Mv^2/P is energy over power, a time; Mv^3/P is a length. This is how the misprint shows itself.
- With $M = F = u = 1$ (so $P = 1$) and $v = 2$, the formulas give $t = 2.5$ and distance $17/6 \approx 2.833$. Stepping the motion numerically (force F below speed u, P/w above) reproduces both values, and agrees with the formulas to better than 10^{-7} for random M, F, u and $v \geq u$; the printed u^2 version disagrees.

Common slips

- Forgetting phase 1 and integrating the constant-power equation from rest. The phase 2 equation only holds once $w \geq u$; starting it from $w = 0$ would mean an infinite force.
- Using $\mathrm{d}w/\mathrm{d}t$ when integrating with respect to x: for distance the acceleration must be written as $w\,\mathrm{d}w/\mathrm{d}x$.
- Writing the phase 1 time as u/F: the acceleration is F/M, not F.

Beyond STEP

The constant-power phase is a first-order differential equation solved by separating variables, and the switch from t to x as independent variable is the same trick used for energy methods in general: $w\,\mathrm{d}w/\mathrm{d}x$ integrates to kinetic energy, which leads to the work–energy theorem and, later, to potentials and conserved quantities. The cap on force at low speed is also how real traction works, since a motor of fixed power cannot

supply unbounded force at rest. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops work, energy and power along these lines.

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